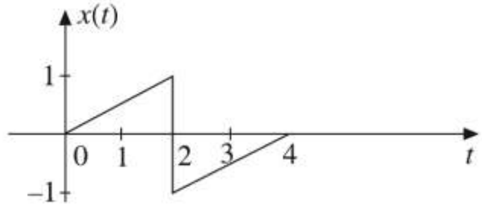
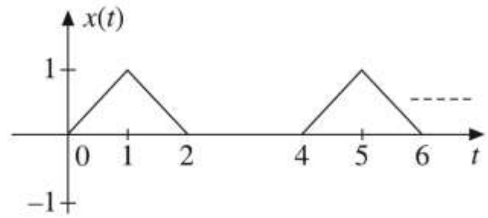
Birla Institute of Technology and Science, Pilani.
Comprehensive Examination: EEE F243/ INSTR F243: Signals and Systems
Marks: 100 AY: 2015-16, Semester: II Date: 5-May-2016, Thursday (FN)
Time: 180 minutes CLOSE BOOK Pages: 02

1 a Express the following signals as sum of singular functions **04**





b Sketch the following signals and determine whether they are periodic or not. Justify your answer. If periodic find the fundamental period. **06**

a. $x(t) = 3u(t) + 2\sin(2t)$
b. $x(t) = 2 + \cos(2\pi t)$
c. $x(t) = \cos(2\pi t + \pi/2)\, u(t)$

c Determine whether time-scaling and time-shifting affects the signal energy? Justify your answer in detail. **04**

2 a Consider a system in figure 2.1. **08**

a. Sketch $X_p(j\omega)$ for $-9\pi \le \omega \le 9\pi$
For the following values of ω_0
1. $\omega_0 = \pi$
2. $\omega_0 = 2\pi$
3. $\omega_0 = 3\pi$
4. $\omega_0 = 5\pi$

Figure 2.1

b. For which of the preceding values of ω_0 (in part (a)), is $x_p(t)$ identical?

b A signal $x(t) = \cos 100\pi t + \cos 250\pi t$ is sampled at sampling frequency f_s and the sampled signal $x_s(t)$ is passed through an ideal low pass filter with bandwidth B. Assuming ideal sampling, sketch the frequency spectrum of the output $y(t)$ of the low pass filter. Also find $y(t)$. **10**

a. For $f_s = 150\ Hz$, and $B = 75\ Hz$, does aliasing occurs?
b. For $f_s = 300\ Hz$, and $B = 150\ Hz$, does aliasing occurs?
c. If the original signal has to be recovered from the sampled signal, what must be the sampling frequency?

3 a Consider a system with input $h(t) = e^{t}u(t)$ and impulse response $x(t) = e^{2t}u(-t)$ and output $y(t)$. **08**

a. Determine $y(t) = x(t) * h(t)$ using graphical method showing sketches of all intermediate steps.
b. Verify your result by using suitable transform.

b If $x(t) = \sqrt{2}(1+j)e^{\frac{j\pi}{4}}e^{(-1+j2\pi)}$. Sketch and label the following: **06**

a. $Re\{x(t)\}$ b. $Im\{x(t)\}$ c. $x(t+2) + x^{*}(t+2)$

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4 a State and prove the integration property of continuous time Fourier transform. Find Fourier transform of unit step signal $u(t)$. **08**

b Find and sketch the frequency domain response of $x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$ **04**

c Find and sketch the frequency domain response of $x[n] = a^{|n|}, \quad |a| < 1.$ **06**

5 a A pressure gauge that can be modeled as an LTI system has a time response to a unit step input given by $(1 - e^{-t} - te^{-t})u(t)$. For a certain input $x(t)$, the output is observed to be $(2 - 3e^{-t} - e^{-3t})u(t)$. For this observed measurement, determine the true pressure input to the gauge as a function of time. (Use Laplace Transform) **06**

b The system function of a causal LTI system is **06**

$$H(s) = \frac{s+1}{s^2 + 2s + 2}.$$

Determine the response $y(t)$ when the input is

$$x(t) = e^{-|t|}, \quad -\infty < t < \infty$$

Plot ROC and comment on stability of system.

c Consider the system S characterized by the differential equation, (use Laplace Transform to) **06**

$$\frac{d^3y(t)}{dt^3} + 6\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 6y(t) = x(t)$$

a. Determine the zero-state response of this system for the input $x(t) = e^{-4t}u(t)$
b. Determine the zero-input response of the system for $t > 0^-$, given that

$$y(0^-) = 1, \quad \left.\frac{dy(t)}{dt}\right|_{t=0^-} = -1, \quad \left.\frac{d^2y(t)}{dt^2}\right|_{t=0^-} = 1.$$

c. Determine the output S when the input is $x(t) = e^{-4t}u(t)$ and the initial conditions are the same as specified in (b).

6 a Given the z-transform pair $x[n] \overset{z}{\leftrightarrow} \frac{z^2}{z^2-16}$, with ROC |z|<4, use the properties to determine the z-transform of the following signals with ROC. **12**

a. $y[n] = x[n-2]$
b. $y[n] = (1/2)^n x[n]$
c. $y[n] = x[-n] * x[n]$
d. $y[n] = nx[n]$
e. $y[n] = x[n+1] + x[n-1]$
f. $y[n] = x[n] * x[n-3]$

b Determine the transfer function and difference equation for impulse response **06**

$$h[n] = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{2}\right)^{n-2} u[n-1]$$

Draw the block diagram representation of the system using Direct Form-I and II.